GROUP WORK PROJECT # 2
GROUP NUMBER: 8215

MScFE 622: Stochastic Modeling

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Team member 2	
Team member 3	

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. Note: You may be required to provide proof of your outreach to non-contributing members upon request.
We tried to reach him several times for him to take part of the group work but he ignored all calls or text message.

STEP 1

1. Introduction

Financial time series data often exhibit non-stationary behavior and varying volatility across different periods. To capture these dynamics, regime-switching models such as the Hamilton Filter are employed to identify distinct market regimes characterized by different statistical properties. This report presents a detailed analysis of financial time series data using a regime-switching model applied to the daily returns of the S&P 500 index (^GSPC) from January 1, 2019, to September 30, 2022.
The study involves data collection, return calculation, exploratory data analysis, volatility estimation, and the implementation of an Expectation-Maximization (EM) algorithm to estimate the parameters of a hidden Markov model with two volatility regimes.

2. Methodology

2.1 Data Collection and Preprocessing

The dataset consists of the adjusted closing prices of the S&P 500 index, downloaded from Yahoo Finance using the `yfinance` Python library. Daily returns are computed as:
R_t = (P_t - P_{t-1}) / P_{t-1}
where P_t is the closing price on day t. Missing values resulting from return computation are removed.

2.2 Exploratory Data Analysis (EDA)

Several statistical methods are employed to understand the return distribution and volatility behavior:
- Time Series Plot: Visualizes daily returns over the selected period.
- Descriptive Statistics: Computes mean, standard deviation, skewness, and kurtosis of returns.
- Histogram and Q-Q Plot: Examines the normality assumption of returns.
- Autocorrelation Function (ACF): Investigates temporal dependencies in the returns.
- Rolling Volatility: A 20-day rolling standard deviation, annualized using sqrt(252), is computed to observe periods of high and low volatility.

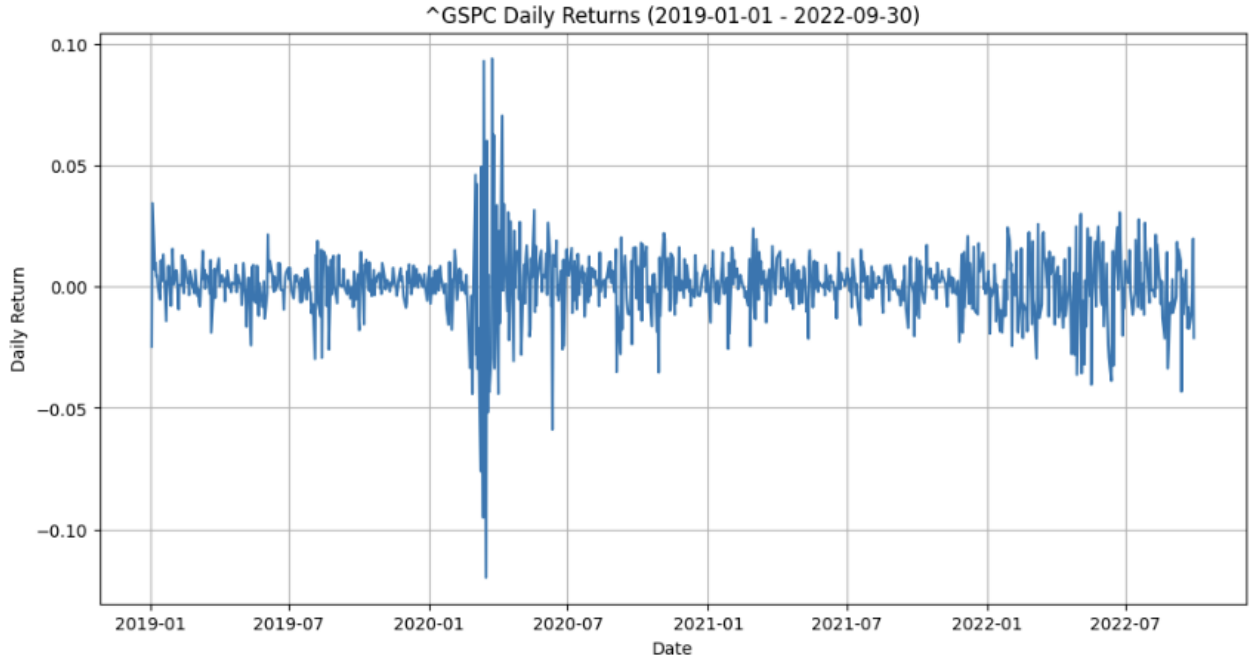
2.3 Regime-Switching Model

A hidden Markov model (HMM) with two regimes is implemented to capture periods of high and low volatility. The model assumes that returns follow a normal distribution with different mean and standard deviation in each regime. The state transition follows a Markov process governed by transition probabilities.

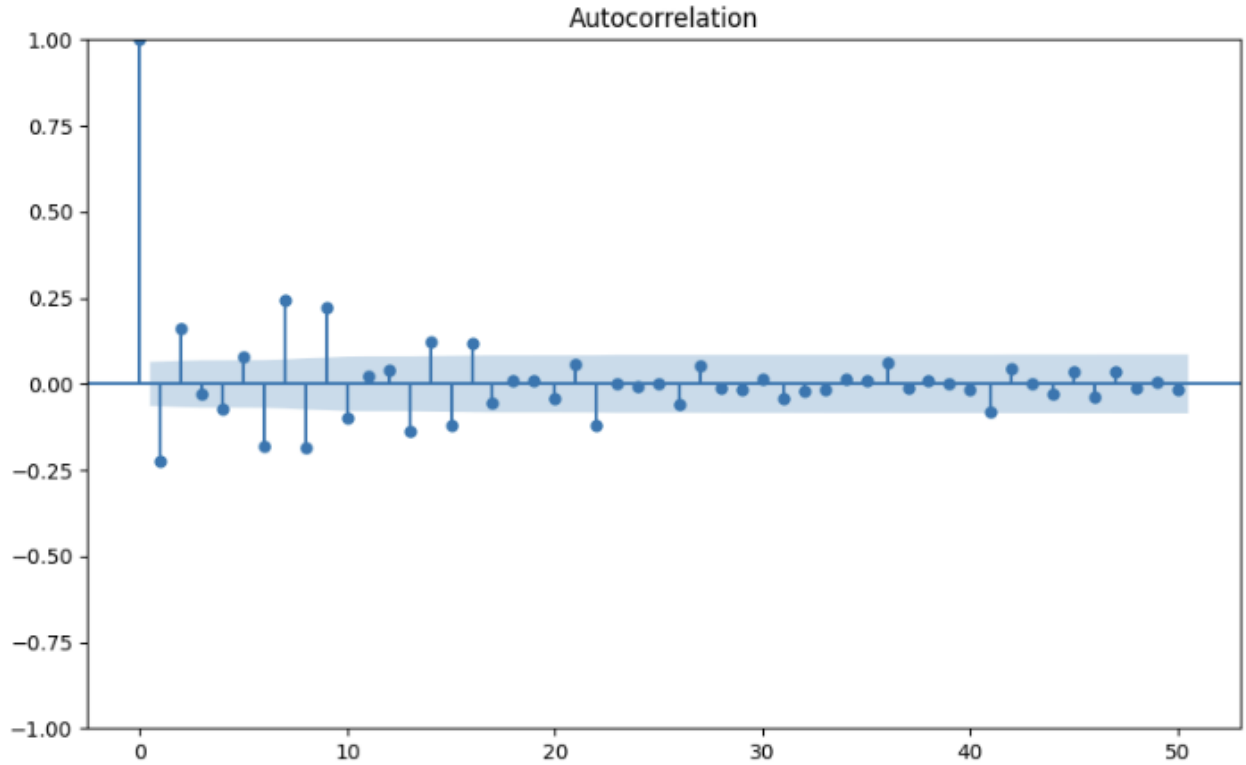
3. Experimental Results

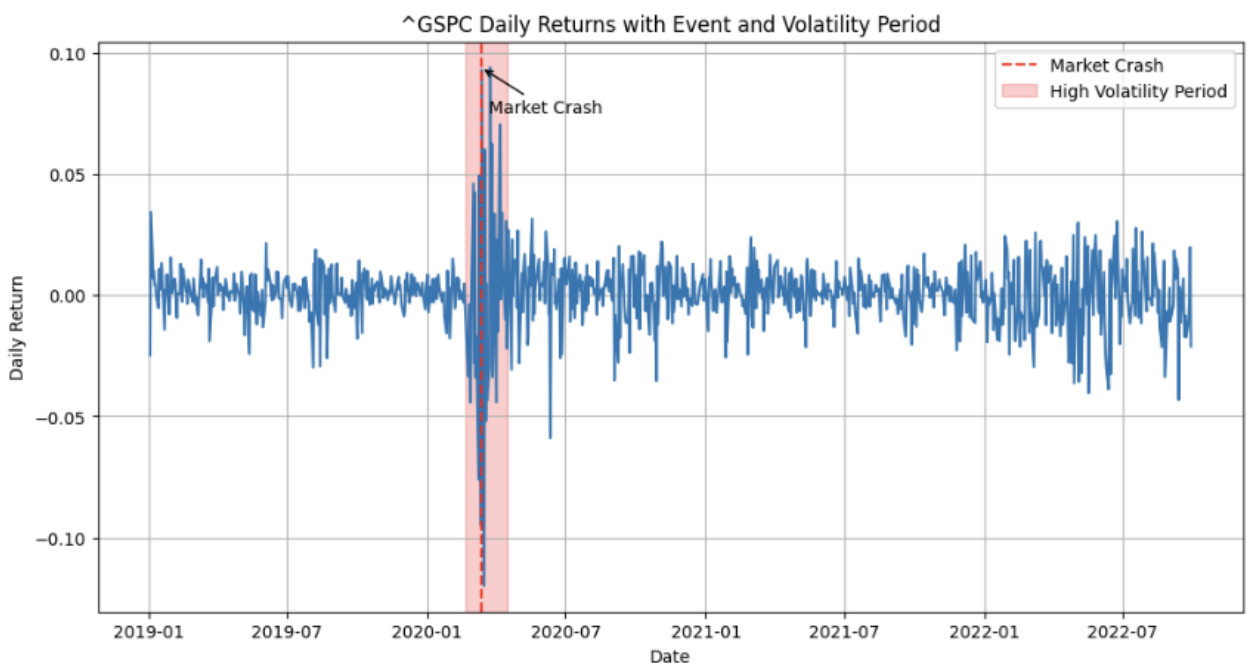
3.1 Exploratory Data Analysis Results

The histogram and Q-Q plot reveal that returns exhibit heavy tails, deviating from normality.
Rolling volatility shows periods of elevated risk, notably during the 2020 COVID-19 market crash.
The ACF plot confirms weak autocorrelation, justifying the use of regime-switching models.
The plots are given below


^GSPC Daily Returns (2019-01-01 - 2022-09-30)
Daily Return
0.10
0.05
0.00
−0.05
−0.10
2019-01
2019-07
2020-01
2020-07
2021-01
2021-07
2022-01
2022-07
Date

20-Day Rolling Volatility of ^GSPC Returns
Volatility
1.0
0.8
0.6
0.4
0.2
2019-01
2019-07
2020-01
2020-07
2021-01
2021-07
2022-01
2022-07
Date





3.2 Regime-Switching Model Results

The estimated means and standard deviations for each regime show distinct market conditions when we consider 2 regimes:

- **Low Volatility Regime**: Lower standard deviation, indicating stable market conditions.
- **High Volatility Regime**: Higher standard deviation, associated with market crashes or uncertainty.

The transition probability matrix suggests persistence within each regime but occasional shifts between them.
The smoothing probabilities confirm that the model successfully identifies major volatility spikes.

- The estimated means and standard deviations for each regime show distinct market conditions when we consider 3 regimes:
  - **Low Volatility Regime:** Lower standard deviation, indicating stable market conditions.
  - **Moderate Volatility Regime:** Intermediate standard deviation, capturing periods of mild fluctuations.
  - **High Volatility Regime:** Higher standard deviation, associated with market crashes or uncertainty.
- The transition probability matrix suggests persistence within each regime but occasional shifts between them.
- The smoothing probabilities confirm that the model successfully identifies major volatility spikes.